

# CacheTest: Smart Cache and `\inln` Fallback

## 1 R

### 1.1 `\runRChunk` — per-chunk cache via MD5

The chunks below are cached individually: editing `r-sum` will only rerun that chunk on the next compile, leaving `r-mean` untouched.

Sum:

```
Error in file(filename, "r", encoding = encoding) :  
  cannot open the connection  
In addition: Warning message:  
In file(filename, "r", encoding = encoding) :  
  cannot open file 'generated/code/hello.R-r-sum.txt': No such file or directory
```

Mean:

```
Error in file(filename, "r", encoding = encoding) :  
  cannot open the connection  
In addition: Warning message:  
In file(filename, "r", encoding = encoding) :  
  cannot open file 'generated/code/hello.R-r-mean.txt': No such file or directory
```

### 1.2 `\runRIncOut` — whole-file cache

Running the full file produces all three outputs:

```
Error in file(filename, "r", encoding = encoding) :  
  cannot open the connection  
In addition: Warning message:  
In file(filename, "r", encoding = encoding) :  
  cannot open file 'code/hello.R': No such file or directory
```

### 1.3 `\inlnR` — backtick inline (Issue #16 fallback)

On first compile there is no cached `.tex` output, so the code must run regardless of the `\ifruncode` flag.

Inline sum: 30

## 2 Python

### 2.1 `\runPythonChunk` — per-chunk cache via MD5

Sum:

```
Error in file(filename, "r", encoding = encoding) :  
  cannot open the connection  
In addition: Warning message:  
In file(filename, "r", encoding = encoding) :  
  cannot open file 'generated/code/hello.py-py-sum.txt': No such file or directory
```

Mean:

```
Error in file(filename, "r", encoding = encoding) :  
  cannot open the connection  
In addition: Warning message:  
In file(filename, "r", encoding = encoding) :  
  cannot open file 'generated/code/hello.py-py-mean.txt': No such file or directory
```

### 2.2 `\runPythonIncOut` — whole-file cache

```
Error in file(filename, "r", encoding = encoding) :  
  cannot open the connection  
In addition: Warning message:  
In file(filename, "r", encoding = encoding) :  
  cannot open file 'code/hello.py': No such file or directory
```

### 2.3 `\inlnPython` — backtick inline (Issue #16 fallback)

Inline sum: Error: unexpected '[' in "print(sum(["

## 3 Julia

### 3.1 `\runJuliaChunk` — per-chunk cache via MD5

Sum:

```
30
```

Mean:

```
6.0
```

### 3.2 `\runJuliaIncOut` — whole-file cache

```
30
6.0
Hello from Julia!
```

### 3.3 `\inlnJulia` — backtick inline (Issue #16 fallback)

Inline sum: 2026-06-13 08:39:34,950 : CRITICAL : client : [julia:65432] Connection refused.  
Is server running?

## 4 Issue #16: `[cache]` package option with missing output

Compiled with `[cache]` the package sets `\runcodefalse`. The Issue #16 fix ensures that if the output `.tex` file does not exist, the code still runs. Delete `generated/` and recompile with `\usepackage[cache,...]{runcode}` to exercise this path.

Fallback: cache-fallback